# SAMIRA ABEDINI




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**GitHub:**

https://github.com/SamiraAbedini


## EDUCATION

**Master of Science in Bioinformatics** October 2024 – now
Saarland University, Saarland, Germany
**Selected Courses:** Neural Networks, Elements of Machine Learning

**Bachelor of Science in Biomedical Engineering** September 2017 – September 2021
(GPA: 16.55/20)
University of Tabriz, Tabriz, Iran
**Selected Courses:** Medical Information Technology, Computer Programming, Mathematics 1 and 2, Biological and Computational Intelligence, Biostatistics, Biophysics, Biochemistry, Medical Physics
**Final Project:** "Designing a Digital Feedforward Dynamic Range Compressor for hearing aids"
(Grade achieved: 20/20)

## WORK EXPERIENCE

**Research Assistant** December 2023 – September 2024
Cognitive Neuroscience Lab., Tabriz University, Tabriz, Iran
- Conducted research on Alzheimer's disease prediction in patients' offspring
- Built and evaluated machine learning models for predictive analysis

**Mathematics teacher (part-time)** April 2020 – September 2023
Hallaj Mathematics Academy, Tabriz, Iran
- Guided high school students in mastering mathematics concepts to excel in university entrance exams.
- Designed and conducted problem-solving sessions to enhance critical thinking and analytical skills.

**Biomedical engineer** December 2020 – January 2023
R&D team, BMGs, Tabriz, Iran
- Analyzed medical data (ECG, EEG) using machine learning techniques
- Programmed touch-screen LCDs for medical devices

## INTERNSHIP

**R&D team, BMG**, Tabriz, Iran August 2020 -- December 2020
**Project**: Designing a piezoelectric energy harvesting system for leadless heart pacemakers, transforming cardiac motion into sustainable electrical energy. (Grade achieved: 20/20)

## PUBLICATION

**Machine Learning – Based Prediction of Early Alzheimer's Biomarkers in High-Risk Offspring** (*under preparation) February 2024 – ongoing
**Authors**: Samira Abedini, Farhad Farkhondeh, Soomayeh Heisiattalab
- Collecting 64 channels resting-state EEG data from two groups of subjects (offspring of Alzheimer's patients, and control group) in both eye-open and eye-close situations
- Preprocessing and creating a dataset from calculated EEG Power Spectrum Density, Relative Power, and Absolute Power of each subject
- Performing statistical analysis and training machine learning models on the dataset to diagnose the early biomarkers of Alzheimer's disease

## COURSES AND CERTIFICATES

**R-Programming** October 2024 – ongoing
DataCamp

**Python for Genomic Data Science (Genomic Data Science Specialization)** October 2023 – January 2024
Coursera, Johns Hopkins University

**Introduction to Genomic Technologies (Genomic Data Science Specialization)** October 2023 – January 2024
Coursera, Johns Hopkins University

**AI for Medical Diagnosis (AI for Medicine Specialization)** June 2023 – September 2023
Coursera, Deeplearning.AI

**Participating in "AI conference, Generative AI Tools"** May 2023
Sharif University of Technology

**Machine Learning Specialization** July 2022 – October 2022
Coursera, Deeplearning.AI

**An Intuitive Introduction to Probability** July 2022
Coursera, Karl Schmedders

**Artificial intelligence school (ML, DL)** January 2022 – March 2022
U-Tech Academy, University of Tehran

## Selected Projects

**ADHD diagnosis from rs-fMRI data using deep learning methods** November 2023 – December 2023
(Using Python) Dataset: ADHD-200 NYU dataset

**Chest X-Ray Medical Diagnosis with Deep Learning** June 2023 – September 2023
(Using Python) Dataset: ChestX-ray8 dataset

**Brain Tumor Auto-Segmentation for Magnetic Resonance Imaging** June 2023 – September 2023
(Using Python) Dataset: Decathlon 10 Challenge dataset

**Object detection project: Detecting Hexbug heads in videos** July 2023
(Using Python) Dataset: AnkiLab(Traco Repository) Hexbugs dataset on GitHub

**Object detection project: Real-time Vehicle detection, tracking, and counting (YOLOv8)** July 2023
(Using Python, YOLOV8) Dataset: Roboflow dataset

## Language Skills

**English**: C1
* TOEFL exam Score: 104 (Reading: 28 Listening: 28 Speaking: 23 Writing: 25) Test date: April 12, 2023
**German**: A2
**Turkish**: C1
**Mother tongue**: Azerbaijani, Persian

## Skills

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|---|---|---|---|
| Python Programming | ●●●●●○ | Matplotlib | ●●●●●○ |
| R Programming | ●●○○○○ | Streamlit | ●●●●○○ |
| C++ | ●●●●○○ | Jupiter Notebook, VS Code | ●●●●●○ |
| TensorFlow | ●●●●○○ | GIT and GitHub | ●●●●●○ |
| Scikitlearn | ●●●●○○ | PyTorch | ●●●○○○ |
| Pandas | ●●●●●○ | MongoDB | ●●●○○○ |
| NumPy | ●●●●○○ | MATLAB | ●●●●●○ |

## Memberships

**Member of Ayaz astronomical society** Ongoing

**Board member of student scientific association of biomedical engineering** October 2019 – June 2020
Faculty of Electrical and Computer Engineering, University of Tabriz

**Member of the executive team of the biomedical engineering conference** September 2019
University of Tabriz

## References

**Associate Professor at the University of Tabriz**
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